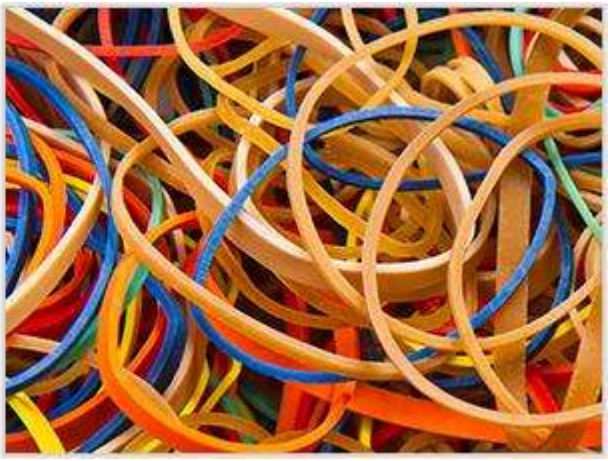
## Description

### Image





### Caption

1. Rubber trees in Kerala, India © M.arunprasad at en.wikipedia - (CC BY-SA 3.0) 2. Rubber bands in different colors. © Bill Ebbesen at en.wikipedia - (CC BY-SA 3.0)

### The material

Natural Rubber was known to the natives of Peru many centuries ago, and is now one of Malaysia's main exports. It made the fortune of Charles Macintosh who, in 1825, devised the rubber-coated waterproof coat the still bears his name. Latex, the sap of the rubber tree, is cross-linked (vulcanized) by heating with sulfur; the amount of the cross-linking determines the properties. It is the most widely used of all elastomers - more than 50% of all produced.

### Compositional summary

(CH2-C(CH3)-CH-CH2)n

## General properties

| | | | | | |
|---|---|---|---|---|---|
| Density | | 930 | - | 970 | kg/m^3 |
| Price | * | 1.17 | - | 1.31 | GBP/kg |
| Date first used | | 1751 | | | |

## Mechanical properties

| | | | | | |
|---|---|---|---|---|---|
| Young's modulus | | 0.0012 | - | 0.0021 | GPa |
| Shear modulus | | 4e-4 | - | 7e-4 | GPa |
| Bulk modulus | * | 1.5 | - | 2 | GPa |
| Poisson's ratio | | 0.499 | - | 0.5 | |
| Yield strength (elastic limit) | | 21 | - | 28 | MPa |
| Tensile strength | | 21 | - | 28 | MPa |
| Compressive strength | * | 25.2 | - | 33.6 | MPa |
| Elongation | | 600 | - | 780 | % strain |

| | | | | | |
|---|---|---|---|---|---|
| Hardness - Vickers | * | 7 | - | 8 | HV |
| Fatigue strength at 10^7 cycles | * | 8.4 | - | 11.2 | MPa |
| Fracture toughness | | 0.109 | - | 0.192 | MPa.m^0.5 |
| Mechanical loss coefficient (tan delta) | | 0.07 | - | 0.13 | |

## Thermal properties

| | | | | | |
|---|---|---|---|---|---|
| Glass temperature | | -78.2 | - | -63.2 | °C |
| Maximum service temperature | | 68.9 | - | 107 | °C |
| Minimum service temperature | | -65.2 | - | -50.2 | °C |
| Thermal conductor or insulator? | | Good insulator | | | |
| Thermal conductivity | | 0.13 | - | 0.16 | W/m.°C |
| Specific heat capacity | | 1.82e3 | - | 1.99e3 | J/kg.°C |
| Thermal expansion coefficient | | 210 | - | 230 | µstrain/°C |

## Electrical properties

| | | | | | |
|---|---|---|---|---|---|
| Electrical conductor or insulator? | | Good insulator | | | |
| Electrical resistivity | | 1e20 | - | 1e23 | µohm.cm |
| Dielectric constant (relative permittivity) | | 2.3 | - | 2.8 | |
| Dissipation factor (dielectric loss tangent) | | 0.001 | - | 0.04 | |
| Dielectric strength (dielectric breakdown) | | 16 | - | 23 | MV/m |

## Optical properties

| | | | | | |
|---|---|---|---|---|---|
| Transparency | | Translucent | | | |

## Critical Materials Risk

| | | | | | |
|---|---|---|---|---|---|
| High critical material risk? | | No | | | |

## Processability

| | | | | | |
|---|---|---|---|---|---|
| Castability | | 4 | - | 5 | |
| Moldability | | 4 | - | 5 | |
| Machinability | | 2 | - | 3 | |
| Weldability | | 1 | | | |

## Durability: water and aqueous solutions

| | | | | | |
|---|---|---|---|---|---|
| Water (fresh) | | Excellent | | | |
| Water (salt) | | Excellent | | | |
| Soils, acidic (peat) | | Excellent | | | |
| Soils, alkaline (clay) | | Excellent | | | |
| Wine | | Excellent | | | |

## Durability: acids

| | |
|---|---|
| Acetic acid (10%) | Excellent |
| Acetic acid (glacial) | Limited use |
| Citric acid (10%) | Excellent |
| Hydrochloric acid (10%) | Limited use |
| Hydrochloric acid (36%) | Limited use |
| Hydrofluoric acid (40%) | Limited use |
| Nitric acid (10%) | Limited use |
| Nitric acid (70%) | Unacceptable |
| Phosphoric acid (10%) | Excellent |
| Phosphoric acid (85%) | Excellent |
| Sulfuric acid (10%) | Excellent |
| Sulfuric acid (70%) | Unacceptable |

## Durability: alkalis

| | |
|---|---|
| Sodium hydroxide (10%) | Excellent |
| Sodium hydroxide (60%) | Limited use |

## Durability: fuels, oils and solvents

| | |
|---|---|
| Amyl acetate | Unacceptable |
| Benzene | Unacceptable |
| Carbon tetrachloride | Unacceptable |
| Chloroform | Unacceptable |
| Crude oil | Unacceptable |
| Diesel oil | Unacceptable |
| Lubricating oil | Unacceptable |
| Paraffin oil (kerosene) | Unacceptable |
| Petrol (gasoline) | Unacceptable |
| Silicone fluids | Excellent |
| Toluene | Unacceptable |
| Turpentine | Unacceptable |
| Vegetable oils (general) | Unacceptable |
| White spirit | Unacceptable |

## Durability: alcohols, aldehydes, ketones

| | |
|---|---|
| Acetaldehyde | Limited use |
| Acetone | Excellent |
| Ethyl alcohol (ethanol) | Excellent |
| Ethylene glycol | |

| | |
|---|---|
| | Excellent |
| Formaldehyde (40%) | Excellent |
| Glycerol | Excellent |
| Methyl alcohol (methanol) | Excellent |

## Durability: halogens and gases

| | |
|---|---|
| Chlorine gas (dry) | Unacceptable |
| Fluorine (gas) | Unacceptable |
| O2 (oxygen gas) | Unacceptable |
| Sulfur dioxide (gas) | Unacceptable |

## Durability: built environments

| | |
|---|---|
| Industrial atmosphere | Excellent |
| Rural atmosphere | Excellent |
| Marine atmosphere | Excellent |
| UV radiation (sunlight) | Poor |

## Durability: flammability

| | |
|---|---|
| Flammability | Highly flammable |

## Durability: thermal environments

| | |
|---|---|
| Tolerance to cryogenic temperatures | Unacceptable |
| Tolerance up to 150 C (302 F) | Acceptable |
| Tolerance up to 250 C (482 F) | Unacceptable |
| Tolerance up to 450 C (842 F) | Unacceptable |
| Tolerance up to 850 C (1562 F) | Unacceptable |
| Tolerance above 850 C (1562 F) | Unacceptable |

## Geo-economic data for principal component

| | | | | |
|---|---|---|---|---|
| Annual world production, principal component | 1.23e7 | | | tonne/yr |

## Primary material production: energy, climate change and water

| | | | | |
|---|---|---|---|---|
| Climate change (CO2-eq) (virgin grade production) | * 2.35 | - | 2.6 | kg/kg |
| Embodied energy (virgin grade production) | * 79.6 | - | 87.8 | MJ/kg |
| Water usage | * 1.5e4 | - | 2e4 | l/kg |

## Material processing: energy

| | | | | |
|---|---|---|---|---|
| Polymer molding (injection, blow), embodied energy | * 16.1 | - | 17.7 | MJ/kg |
| Machining, grinding, embodied energy | * 22.5 | - | 27.3 | MJ/kg |

## Material processing: climate change

| | | | | |
|---|---|---|---|---|
| Polymer molding (injection, blow), climate change (CO2-eq) | * 1.05 | - | 1.16 | kg/kg |
| Machining, grinding, climate change (CO2-eq) | * 1.47 | - | 1.79 | kg/kg |

## Recycling and end of life

| | | | | |
|---|---|---|---|---|
| Recycle | ✗ | | | |
| Functional recycle | ✗ | | | |
| Combust for energy recovery | ✓ | | | |
| Heat of combustion (net) | * 42.5 | - | 44.6 | MJ/kg |
| Combustion CO2 | * 3.15 | - | 3.31 | kg/kg |
| Landfill | ✓ | | | |
| Biodegrade | ✗ | | | |
| Toxicity rating | Non-toxic | | | |
| A renewable resource? | ✓ | | | |

### Environmental notes

Natural rubber is a biopolymer. Once vulcanized, natural rubbers are thermosets, and thus cannot be recycled. Their disposal creates an environmental problem.

### Recycle mark




## Supporting information

### Design guidelines

Natural rubber is an excellent, cheap, general-purpose elastomer with large stretch capacity and useful properties from -50 C to 115 C, but with poor oil, oxidation, ozone and UV resistance. It has low hysteresis - and is thus very bouncy.

### Typical uses

Gloves, Car tires, seals, belts, anti-vibration mounts, electrical insulation, tubing, rubber lining pipes and pumps.

## Links

ProcessUniverse

Producers

Reference